UDB LAB ASSIGNMENT – 8

NAME :
REGISTRATION NO:AU/2023/0009351

ROLL NO. :UG/02/BTCSECSF/2023/012

SUBJECT : Unstructured Database Lab SEC: E

COLLECTION 1: **flights**
Schema
{

flight_id: Number,
airline: String,
source: String,
destination: String,
price: Number,
duration: Number
}
Table

flight_id	airline	source	destination	price	duration
1	Indigo	Delhi	Mumbai	5000	2
2	AirIndia	Kolkata	Chennai	6500	3
3	SpiceJet	Mumbai	Bangalore	4000	1.5
4	Vistara	Delhi	Chennai	7000	3
5	GoFirst	Mumbai	Delhi	4800	2
6	Indigo	Bangalore	Kolkata	5500	2.5

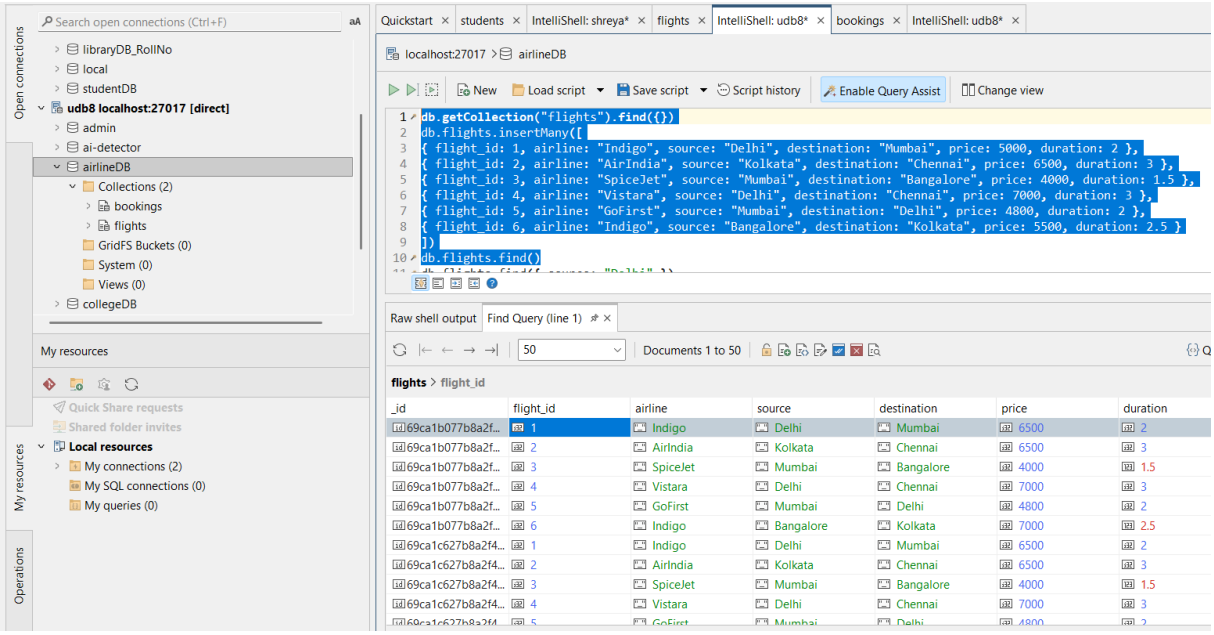
COLLECTION 2: **bookings**
Schema
{
booking_id: Number,
passenger_name: String,
flight_id: Number,
city: String,
seats: Number,
total_fare: Number
}
Table

booking_id	passenger_name	flight_id	city	seats	total_fare
101	Aarav	1	Delhi	1	5000
102	Meera	2	Kolkata	2	13000
103	Rohan	3	Mumbai	1	4000
104	Sneha	4	Delhi	3	21000

booking_id	passenger_name	flight_id	city	seats	total_fare
105	Kunal	5	Mumbai	2	9600
106	Priya	6	Bangalore	1	5500

QUESTIONS

Q1. Display all flights.



Q2. Display passenger_name and city from bookings.

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Query Assist | Change view

```
1  db.getCollection("bookings").find({})
2  db.bookings.insertMany([
3  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats: 1, total_fare: 5000 },
4  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats: 2, total_fare: 13000 },
5  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats: 1, total_fare: 4000 },
6  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats: 3, total_fare: 21000 },
7  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats: 2, total_fare: 9600 },
8  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore", seats: 1, total_fare: 5500 }
9  ])
10 db.bookings.find({}, { passenger_name: 1, city: 1, _id: 0 })
11 db.bookings.find({ seats: 1 })
12 db.bookings.find().sort({ total_fare: -1 }).limit(2)
13 db.bookings.find({ seats: { $gte: 1, $lte: 2 } })
14 db.bookings.deleteMany({ seats: { $gt: 2 } })
15 db.bookings.aggregate([
16   { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
```

Raw shell output | Find Query (line 10)

50 | Documents 1 to 30

bookings > city

passenger_name	city
Aarav	Delhi
Meera	Kolkata
Rohan	Mumbai
Kunal	Mumbai
Priya	Bangalore
Aarav	Delhi

1 document selected

Q3. Find flights where source is Delhi.

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Query Assist | Change view

```
4  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Chennai", price: 6500, duration: 3 },
5  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bangalore", price: 4000, duration: 1.5 },
6  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai", price: 7000, duration: 3 },
7  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi", price: 4800, duration: 2 },
8  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolkata", price: 5500, duration: 2.5 }
9  ])
10 db.flights.find()
11 db.flights.find({ source: "Delhi" })
12 db.flights.find().sort({ price: 1 })
13 db.flights.find({ price: { $gt: 5000 } })
14 db.flights.find({ destination: { $ne: "Chennai" } })
15 db.flights.updateMany(
16   { airline: "Indigo" },
17   { $inc: { price: 500 } }
18 )
19 db.flights.aggregate([
20   { $group: { _id: "$airline", avg_price: { $avg: "$price" } } }
21 ])
22 db.flights.aggregate([
```

Raw shell output | Find Query (line 11)

50 | Documents 1 to 22 | Query (

flights > flight_id

_id	flight_id	airline	source	destination	price	duration
69ca1b077b8a2f...	1	Indigo	Delhi	Mumbai	6500	2
69ca1b077b8a2f...	4	Vistara	Delhi	Chennai	7000	3

Q4. Find bookings where seats = 1.

Quickstart × students × IntelliShell: shreya* × flights × IntelliShell: udb8* × bookings × IntelliShell: udb8* ×

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Query Assist | Change view

```
db.getCollection("bookings").find({})
db.bookings.insertMany([
{ booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats: 1, total_fare: 5000 },
{ booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats: 2, total_fare: 13000 },
{ booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats: 1, total_fare: 4000 },
{ booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats: 3, total_fare: 21000 },
{ booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats: 2, total_fare: 9600 },
{ booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore", seats: 1, total_fare: 5500 }
])
db.bookings.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.bookings.find({ seats: 1 })
db.bookings.find().sort({ total_fare: -1 }).limit(2)
db.bookings.find({ seats: { $gte: 1, $lte: 2 } })
db.bookings.deleteMany({ seats: { $gt: 2 } })
db.bookings.aggregate([
```

Raw shell output | Find Query (line 11)

50 | Documents 1 to 18 | Query Co

bookings > booking_id

_id	booking_id	passenger_name	flight_id	city	seats	total_fare
69ca1b221be3e0...	101	Aarav	1	Delhi	1	5000
69ca1b221be3e0...	103	Rohan	3	Mumbai	1	4000
69ca1b221be3e0...	106	Priya	6	Bangalore	1	5500

Q5. Sort flights by price in ascending order.

Quickstart × students × IntelliShell: shreya* × flights × IntelliShell: udb8* × bookings × IntelliShell: udb8* ×

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Query Assist | Change view

```
{ flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Chennai", price: 6500, duration: 3 },
{ flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bangalore", price: 4000, duration: 1.5 },
{ flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai", price: 7000, duration: 3 },
{ flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi", price: 4800, duration: 2 },
{ flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolkata", price: 5500, duration: 2.5 }
])
db.flights.find()
db.flights.find({ source: "Delhi" })
db.flights.find().sort({ price: 1 })
db.flights.find({ price: { $gt: 5000 } })
db.flights.find({ destination: { $ne: "Chennai" } })
db.flights.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
```

Raw shell output | Find Query (line 12)

50 | Documents 1 to 50 | Query Cod

flights > flight_id

_id	flight_id	airline	source	destination	price	duration
69ca1e587b8a2f4...	3	SpiceJet	Mumbai	Bangalore	4000	1.5
69ca1b077b8a2f...	5	GoFirst	Mumbai	Delhi	4800	2

Q6. Display top 2 bookings with highest total_fare.

Quickstart × | students × | IntelliShell: shreya* × | flights × | IntelliShell: udb8* × | bookings × | IntelliShell: udb8* ×

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Query Assist | Change view

```
1  db.getCollection("bookings").find({})
2  db.bookings.insertMany([
3  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats: 1, total_fare: 5000 },
4  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats: 2, total_fare: 13000 },
5  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats: 1, total_fare: 4000 },
6  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats: 3, total_fare: 21000 },
7  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats: 2, total_fare: 9600 },
8  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore", seats: 1, total_fare: 5500 }
9  ])
10 db.bookings.find({}, { passenger_name: 1, city: 1, _id: 0 })
11 db.bookings.find({ seats: 1 })
12 db.bookings.find().sort({ total_fare: -1 }).limit(2)
13 db.bookings.find({ seats: { $gte: 1, $lte: 2 } })
14 db.bookings.deleteMany({ seats: { $gt: 2 } })
15 db.bookings.aggregate([
16   { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
```

Raw shell output | Find Query (line 12)

50 | Documents 1 to 2 | Query C

bookings > booking_id

_id	booking_id	passenger_name	flight_id	city	seats	total_fare
69ca1b381be3e0...	102	Meera	2	Kolkata	2	13000
69ca1b221be3e0...	102	Meera	2	Kolkata	2	13000

Q7. Find flights where price is greater than 5000.

Quickstart × | students × | IntelliShell: shreya* × | flights × | IntelliShell: udb8* × | bookings × | IntelliShell: udb8* ×

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Query Assist | Change view

```
4  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Chennai", price: 6500, duration: 3 },
5  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bangalore", price: 4000, duration: 1.5 },
6  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai", price: 7000, duration: 3 },
7  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi", price: 4800, duration: 2 },
8  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolkata", price: 5500, duration: 2.5 }
9  ])
10 db.flights.find()
11 db.flights.find({ source: "Delhi" })
12 db.flights.find().sort({ price: 1 })
13 db.flights.find({ price: { $gt: 5000 } })
14 db.flights.find({ destination: { $ne: "Chennai" } })
15 db.flights.updateMany(
16   { airline: "Indigo" },
17   { $inc: { price: 500 } }
18 )
```

Raw shell output | Find Query (line 13)

50 | Documents 1 to 43

flights > flight_id

_id	flight_id	airline	source	destination	price	duration
69ca1b077b8a2f...	1	Indigo	Delhi	Mumbai	6500	2
69ca1b077b8a2f...	2	AirIndia	Kolkata	Chennai	6500	3
69ca1b077b8a2f...	4	Vistara	Delhi	Chennai	7000	3
69ca1b077b8a2f...	6	Indigo	Bangalore	Kolkata	7000	2.5
69ca1c627b8a2f4...	1	Indigo	Delhi	Mumbai	6500	2
69ca1c627b8a2f4...	2	AirIndia	Kolkata	Chennai	6500	3
69ca1c627b8a2f4...	4	Vistara	Delhi	Chennai	7000	3

1 document selected

Q8. Display flights where destination is not Chennai.

```
db.flights.find({ price: { $gt: 5000 } })
db.flights.find({ destination: { $ne: "Chennai" } })
db.flights.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
```

shell output | Find Query (line 14)

50 | Documents 1 to 44

ghts > flight_id

	flight_id	airline	source	destination	price	duration
69ca1b077b8a2f...	1	Indigo	Delhi	Mumbai	6500	2
69ca1b077b8a2f...	3	SpiceJet	Mumbai	Bangalore	4000	1.5
69ca1b077b8a2f...	5	GoFirst	Mumbai	Delhi	4800	2
69ca1b077b8a2f...	6	Indigo	Bangalore	Kolkata	7000	2.5
69ca1c627b8a2f4...	1	Indigo	Delhi	Mumbai	6500	2
69ca1c627b8a2f4...	3	SpiceJet	Mumbai	Bangalore	4000	1.5

Q9. Find bookings where seats are between 1 and 2.

Quickstart | students | IntelliShell: shreya* | flights | IntelliShell: udb8* | bookings | IntelliShell: udb8*

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Query Assist | Change view

```
1  db.getCollection("bookings").find({})
2  db.bookings.insertMany([
3  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats: 1, total_fare: 5000 },
4  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats: 2, total_fare: 13000 },
5  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats: 1, total_fare: 4000 },
6  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats: 3, total_fare: 21000 },
7  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats: 2, total_fare: 9600 },
8  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore", seats: 1, total_fare: 5500 }
9  ])
10 db.bookings.find({}, { passenger_name: 1, city: 1, _id: 0 })
11 db.bookings.find({ seats: 1 })
12 db.bookings.find().sort({ total_fare: -1 }).limit(2)
13 db.bookings.find({ seats: { $gte: 1, $lte: 2 } })
14 db.bookings.deleteMany({ seats: { $gt: 2 } })
15 db.bookings.aggregate([
16   { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
```

Raw shell output | Find Query (line 13)

50 | Documents 1 to 30

bookings > booking_id

_id	booking_id	passenger_name	flight_id	city	seats	total_fare
69ca1b221be3e0...	101	Aarav	1	Delhi	1	5000
69ca1b221be3e0...	102	Meera	2	Kolkata	2	13000
69ca1b221be3e0...	103	Rohan	3	Mumbai	1	4000
69ca1b221be3e0...	105	Kunal	5	Mumbai	2	9600
69ca1b221be3e0...	106	Priya	6	Bangalore	1	5500
69ca1b381be3e0...	101	Aarav	1	Delhi	1	5000

Q10. Update price by +500 for flights of airline "Indigo".

```
db.flights.find({ destination: { $ne: "Chennai" } })
db.flights.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
db.flights.aggregate([
  { $group: { _id: "$airline", avg_price: { $avg: "$price" } } }
])
db.flights.aggregate([
  { $group: { _id: "$airline", avg_price: { $avg: "$price" } } },
```

Raw shell output | Shell Output (Documents)

50 | Documents 1 to 1

Result > insertedId

acknowledged	insertedId	matchedCount	modifiedCount	upsertedCount
true	null	22.0	22.0	0.0

Q11. Delete bookings where seats are greater than 2.

```
db.bookings.deleteMany({ seats: { $gt: 2 } })
db.bookings.aggregate([
```

Raw shell output | Shell Output (Documents)

50 | Documents 1 to 1

Result > deletedCount

acknowledged	deletedCount
true	0.0

Q12. Find total fare collected city-wise. *(Aggregation)*

```
14 db.bookings.deleteMany({ seats: { $gt: 2 } })
15 db.bookings.aggregate([
16   { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
17 ])
```

Raw shell output | Aggregate Query (line 15)

50 | Documents 1 to 4

bookings > total_fare

_id	total_fare
Mumbai	95200
Delhi	35000
Kolkata	91000
Bangalore	38500

Q13. Find average flight price per airline. *(Aggregation)*

```
19 db.flights.aggregate([
20   { $group: { _id: "$airline", avg_price: { $avg: "$price" } } }
21 ])
22 db.flights.aggregate([
23   { $group: { _id: "$airline", avg_price: { $avg: "$price" } } },
24   { $match: { avg_price: { $gt: 5500 } } }
25 ])
```

aw shell output | Aggregate Query (line 19)

50 | Documents 1 to 5

lights > avg_price

id	avg_price
GoFirst	4800.0
SpiceJet	4000.0
Indigo	6977.272727272...
AirIndia	6500.0
Vistara	7000.0

Q14. Display airlines where average price is greater than 5500. *(Aggregation + filter)*

Quickstart × | students × | IntelliShell: shreya* × | flights × | IntelliShell: udb8* × | bookings

localhost:27017 > airlineDB

New | Load script | Save script | Script history | Enable Que

```
11 db.flights.find({ source: "Delhi" })
12 db.flights.find().sort({ price: 1 })
13 db.flights.find({ price: { $gt: 5000 } })
14 db.flights.find({ destination: { $ne: "Chennai" } })
15 db.flights.updateMany(
16   { airline: "Indigo" },
17   { $inc: { price: 500 } }
18 )
19 db.flights.aggregate([
20   { $group: { _id: "$airline", avg_price: { $avg: "$price" } } }
21 ])
22 db.flights.aggregate([
23   { $group: { _id: "$airline", avg_price: { $avg: "$price" } } },
24   { $match: { avg_price: { $gt: 5500 } } }
25 ])
```

Raw shell output | Aggregate Query (line 22)

50 | Documents 1 to 3

flights > avg_price

_id	avg_price
Vistara	7000.0
AirIndia	6500.0
Indigo	6977.272727272...

Q15. Find maximum total_fare among bookings for each city. *(Aggregation)*

```
26 db.bookings.aggregate([
27   { $group: { _id: "$city", max_fare: { $max: "$total_fare" } } }
28 ])
```

Raw shell output | Aggregate Query (line 26)

50 | Documents 1 to 4

bookings > max_fare

_id	max_fare
Mumbai	9600
Delhi	5000
Kolkata	13000
Bangalore	5500